

Alex Liu

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Education

University of Florida (UF) - Gainesville, FL

Expected Graduation: **May 2028**

Bachelor of Science in Computer Science, Minor in Geography

GPA: 3.80/4.0

- **Relevant Coursework:** Computer Organization, Linear Algebra, Data Structures and Algorithms, Programming Fundamentals 2, Discrete Structures, Calculus 3

Skills

Languages: Python, C++, C, Java, JavaScript, TypeScript, Go, C#, SQL, RISC-V Assembly, Bash, HTML/CSS

Framework & Library: React.js, FastAPI, Flask, Node.js, OpenCV, MediaPipe, SQLAlchemy, NumPy, Pandas, TailwindCSS

Developer Tools: Git, GitHub, Docker, GitHub Actions (CI/CD), Linux/Unix, Postman, Vite, PyAutoGUI

Platforms & Databases: AWS (Lambda/ECS/S3/RDS), PostgreSQL, Vercel, Railway, Microservices, REST APIs, JWT

Projects

Geography Dashboard, *React, AWS, Leaflet, FastAPI, Vite, TypeScript, PostgreSQL, Docker, Gemini AI*

Nov 2025

- Engineered a **full-stack** geospatial platform using a **microservices architecture** to enable **distributed deployment**, decoupling core GIS processing from **AI services** while ensuring **cross-cloud** observability by propagating X-Request-ID headers across FastAPI and AWS services.
- Architected a cloud-native AI engine on **AWS**, utilizing **HTTP APIs** to process structured spatial data (DBSCAN clusters, density grids) and generate interpretive summaries via Gemini API.
- Systematized a **CI/CD pipeline** via GitHub Actions to automate deployment, ensuring consistent configuration and **fault isolation** across cross-cloud environments while implementing database-backed **rate limiting**.
- Developed high-performance **REST APIs** with **FastAPI** and **SQLAlchemy** to process structured geospatial datasets(CSV/GeoJSON), transforming raw spatial results into interpretive analytical summaries for city-scale data.

UF Health SmartScribe, *HTML, CSS, JavaScript, Supabase, Gemini AI*

Oct 2025

- Engineered a real-time **full-stack** medical documentation prototype within **24 hours**, integrating Deepgram and Gemini AI API to transcribe consultations and auto-generate structured SOAP notes.
- Architected a secure backend pipeline using **Supabase** to manage patient records, automating clinical workflows to reduce documentation time by **60%**.
- Implemented a speaker-aware AI extraction engine processing **60+ words/minute** with **sub-5-second** latency, prioritizing immediate physician utility over administrative overhead

Grade Track, *Flask, React.js, SQLAlchemy, PostgreSQL, Docker*

Oct 2025

- Engineered a **full-stack** analytics dashboard to automate academic performance tracking, utilizing a **CSV ingestion pipeline** and **Recharts** to transform raw data into real-time visual insights.
- Orchestrated containerized deployment with **Docker**, achieving an **85%** increase in operational efficiency.

Ability Bridge, *Python, OpenCV, MediaPipe, PyAutoGUI*

Sep 2025

- Engineered a hands-free accessibility interface utilizing **OpenCV**, processing **14,000+** facial landmarks/second to enable precise **real-time cursor control** and **Morse-code** text entry.
- Optimized signal processing via **exponential smoothing** and **hysteresis-based calibration** to achieve **30 FPS**, effectively eliminating cursor jitter and false triggers across diverse lighting conditions.

Experiences & Involvements

SASE Intern - UF Society of Asian Scientists & Engineers (SASE)

Aug 2025 - Dec 2025

- **Coordinated** professional events for **150+ attendees**, managing logistics and cross-functional team communication to enhance member career readiness.
- **Collaborated** with a leadership board of **100+** to streamline organizational programming and mentorship initiatives.

Event Committee Member - UF Chinese American Student Association (CASA)

Aug 2025 - Oct 2025

- **Spearheaded** a large-scale cultural festival for **300+ participants**, overseeing program design and day-of execution.
- **Curated** and managed cultural activities to increase student engagement and celebrate heritage for the UF community.

Instructor - Logic Lab

June 2022 – July 2022

- Led and instructed a class of **30+** students in math and chess, designing lessons with **4** other team members.